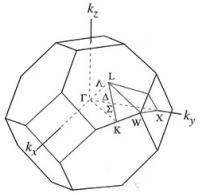
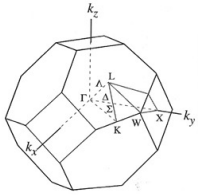
Continuity Equation



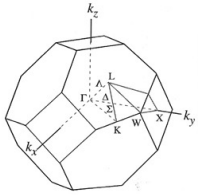
What We Have Learned So Far?

- How to calculate current?
  - ➢How to calculate carrier density
    - ✓State of density
    - ✓Fermi-Dirac Distribution
  - ➢How to calculate velocity?
    - ✓Drift
    - ✓Diffuse
- It seems we still need a couple of more things (or more equations) to calculate all these quantities.

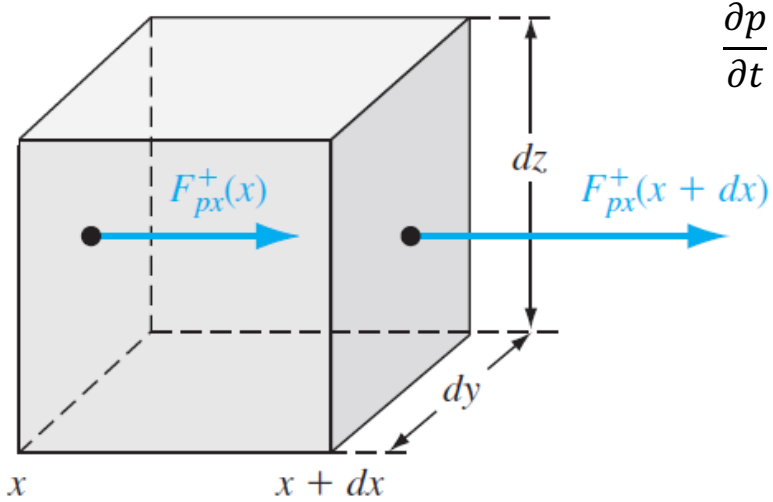


What Do We Miss So Far?

- Electron concentration: n(x)
- Hole concentration: p(x)
- Electric Field: E(x)
- How we connect all these together?



Continuity Equation (so far)



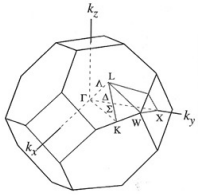
$$\frac{\partial p}{\partial t}dxdydz = \left[F_{px}^{+}(x) - F_{px}^{+}(x+dx)\right]dydz = -\frac{\partial F_{px}^{+}}{\partial x}dxdydz$$

$$\frac{\partial p}{\partial t}dxdydz = -\frac{\partial F_{px}^{+}}{\partial x}dxdydz$$

$$\frac{\partial p}{\partial t} = -\frac{\partial F_{px}^{+}}{\partial x}$$

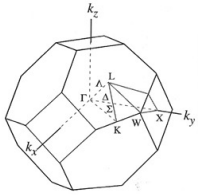
$$\frac{\partial p}{\partial t} = -\frac{1}{e}\frac{\partial J_p}{\partial x}$$

$$\frac{\partial n}{\partial t} = +\frac{1}{e}\frac{\partial J_n}{\partial x}$$



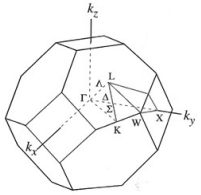
Continuity Equation (3D)

- $\frac{\partial p}{\partial t} = -\frac{1}{e}\left(\frac{\partial J_p}{\partial x} + \frac{\partial J_p}{\partial y} + \frac{\partial J_p}{\partial z}\right)$
- $\frac{\partial n}{\partial t} = +\frac{1}{e}\left(\frac{\partial J_n}{\partial x} + \frac{\partial J_n}{\partial y} + \frac{\partial J_n}{\partial z}\right)$
- $\frac{\partial p}{\partial t} = -\frac{1}{e}(\nabla \cdot J_p)$
- $\frac{\partial n}{\partial t} = +\frac{1}{e}(\nabla \cdot J_n)$



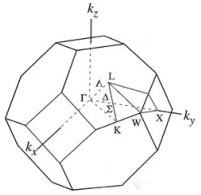
What Else?

- Electric Field E(x)
- Poisson's Equation
  - $\nabla \cdot D = q(p - n + N_d^+ - N_a^-)$
  - $D = \varepsilon E(x)$



Transport Equations in 3D

- $\frac{\partial p}{\partial t} = -\frac{1}{q}\nabla \cdot J_q$
- $\frac{\partial n}{\partial t} = \frac{1}{q}\nabla \cdot J_n$
- $J_p = ep\mu_p \vec{E} - eD_p \nabla p$
- $J_n = en\mu_n \vec{E} + eD_n \nabla n$
- $\nabla \cdot D = q(p - n + N_d^+ - N_a^-)$
- What is the assumption we use to derive this equation?

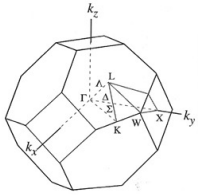


Generation and Recombination

Baile Chen

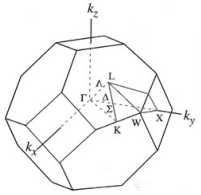
School of Information Science and Technology

Shanghai Tech University



Today We Turn the Light ON Semiconductor

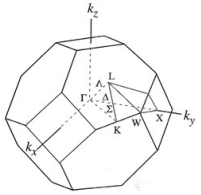
- Before we do, recall that the light OFF, the number of “free” carriers in a sample are just given by:
  - Thermal generation
  - Charge neutrality
  - $n_0 + N_a^- = p_0 + N_d^+$
  - The only unknown is the Fermi Level.



Band to Band Generation and Recombination

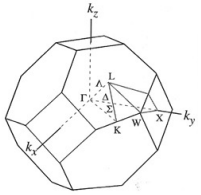


Figure 6.1 | Electron–hole generation and recombination.



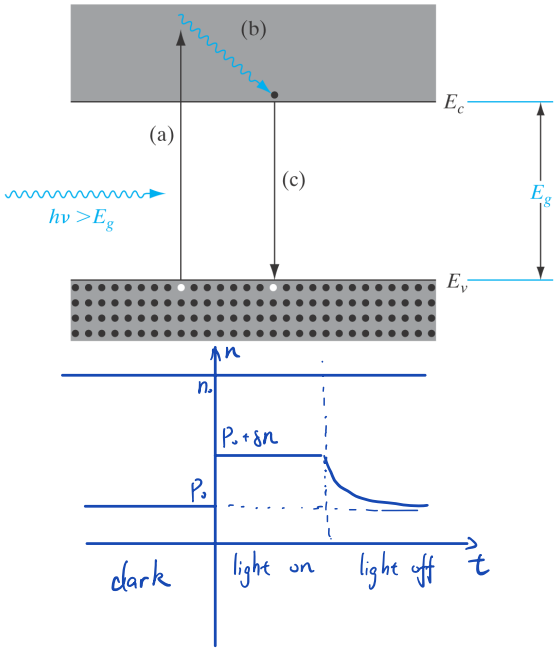
Semiconductor in Equilibrium

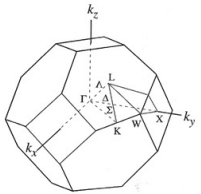
- Generation Rate:
  - Unit: #/cm^3 · s^{-1}
  - $G_{n0} = G_{p0} = A$
- Recombination Rate:
  - Unit: #/cm^3 · s^{-1}
  - $R_{n0} = R_{p0} = B(n_0 \cdot p_0)$
- $\frac{dn(t)}{dt} = \frac{dp(t)}{dt} = G_{n0} - R_{n0} = G_{p0} - R_{p0}$
- In Equilibrium:
  - What happened?
  - What is relation between A and B?



Now We Turn the Light ON Semiconductor

- What is the condition for light absorption?





n type $n_o = N_D \gg p_o$

$n = n_o + \delta n \quad p = p_o + \delta p$

$$\frac{dn}{dt} = A - Bnp$$

$$= B(n_i^2 - np)$$

$$= B[n_i^2 - (n_o + \delta n)(p_o + \delta p)]$$

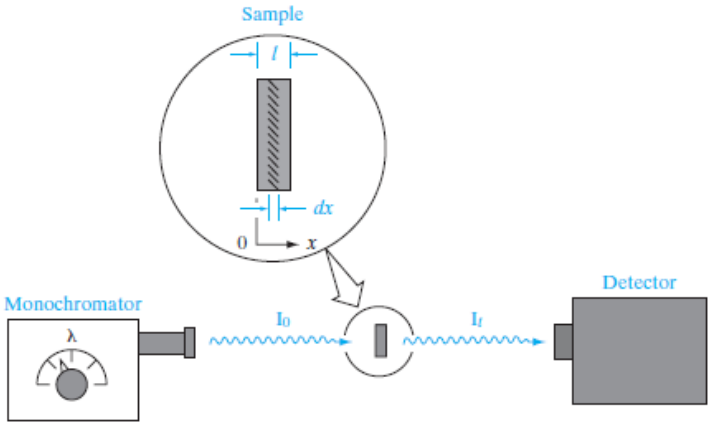
$$\Rightarrow \frac{d\delta n}{dt} = -Bn_o \delta p \qquad n = n_o + \delta n = n_o + \delta p$$

$$\Rightarrow \frac{d\delta p}{dt} = -Bn_o \delta p \qquad \frac{d\delta n}{dt} = -Bn_o \delta n$$

$$\delta p = \delta p_{(0)} e^{-\frac{t}{\tau}} \quad \tau = \frac{1}{Bn_o}$$

Light Absorption

- Optical Absorption experiment

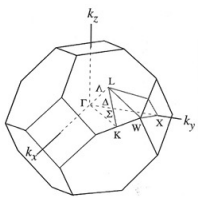


$$\frac{dI(x)}{dx} = -\alpha I(x)$$

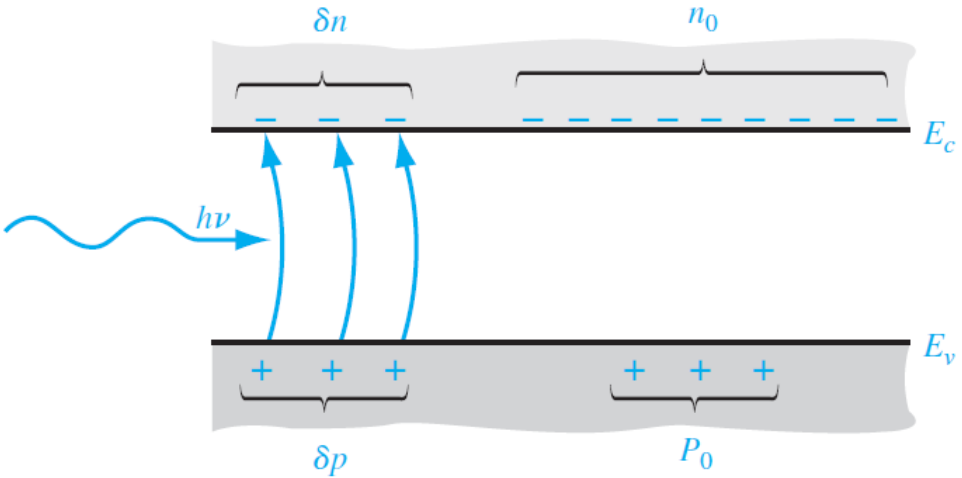
α is the absorption coefficient

$$I(x) = I_0 e^{-\alpha x}$$

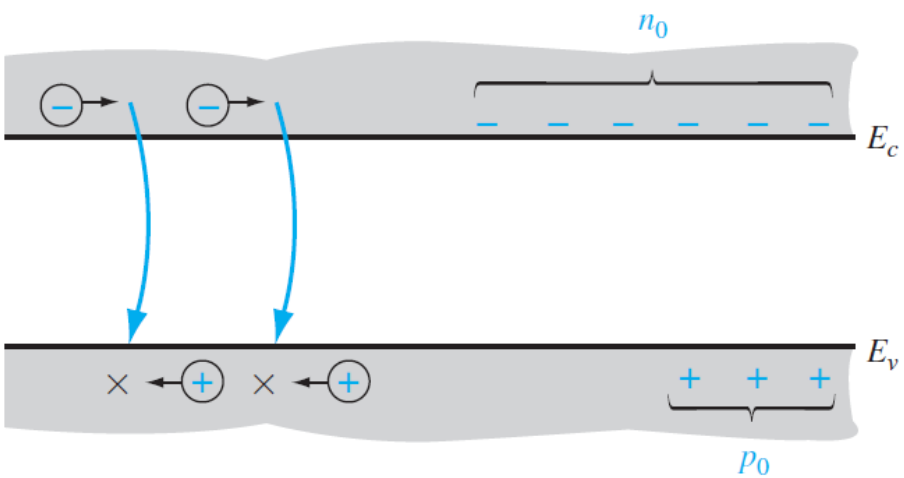
semiconductors absorb photons much more efficiently at energies greater than the band gap



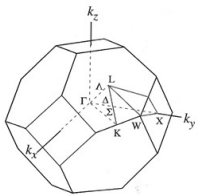
Excess Carrier Generation and Recombination



Generation of Excess electron and hole by optical absorption

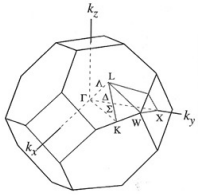


Recombination of Excess electron and hole reestablishing thermal equiblirium



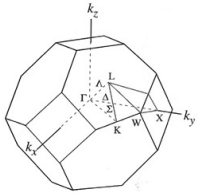
Optical Absorption

- Light absorbed created <u>excess EHPs</u>
  - excess with respect to what?
  - $n(t) = n_0 + \delta n(t)$
  - $p(t) = p_0 + \delta p(t)$
- How long do excess EHPs “live” before they recombine?
- Direct EHP recombination occurs spontaneously, emitting a photon of energy of ??



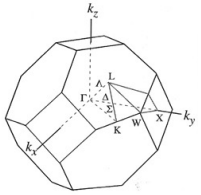
Rate Equation

- $\frac{dn(t)}{dt} = G_n - R_n = A - Bn \cdot p$
- $A = B(n_0 \cdot p_0) = Bn_i^2$
- $\frac{dn(t)}{dt} = Bn_i^2 - Bn \cdot p = B\left(n_i^2 - n \cdot p\right)$
- $n(t) = n_0 + \delta n(t)$
- $\frac{d\delta n(t)}{dt} = B\left(n_i^2 - (n_0 + \delta n(t)) \cdot (p_0 + \delta p(t))\right)$
- $\frac{d\delta n(t)}{dt} = -B\delta n(t)((n_0 + p_0) + \delta n(t))$



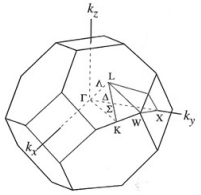
Low level Injection

- How do excess EHPs decay?
  - Assume p-type sample (p_0 >> n_0) so electrons are in minority
  - Will majority carriers (holes) be disturbed much?
  - What about minority carriers (electrons)?
- $\frac{d\delta n(t)}{dt} = -B\delta \mathrm{n(t)}\, p_0$
- $\delta n(t) = \delta n(0) e^{-Bp_0 t} = \delta n(0) e^{-t/\tau_p}$
- τ_p: excess carrier lifetime

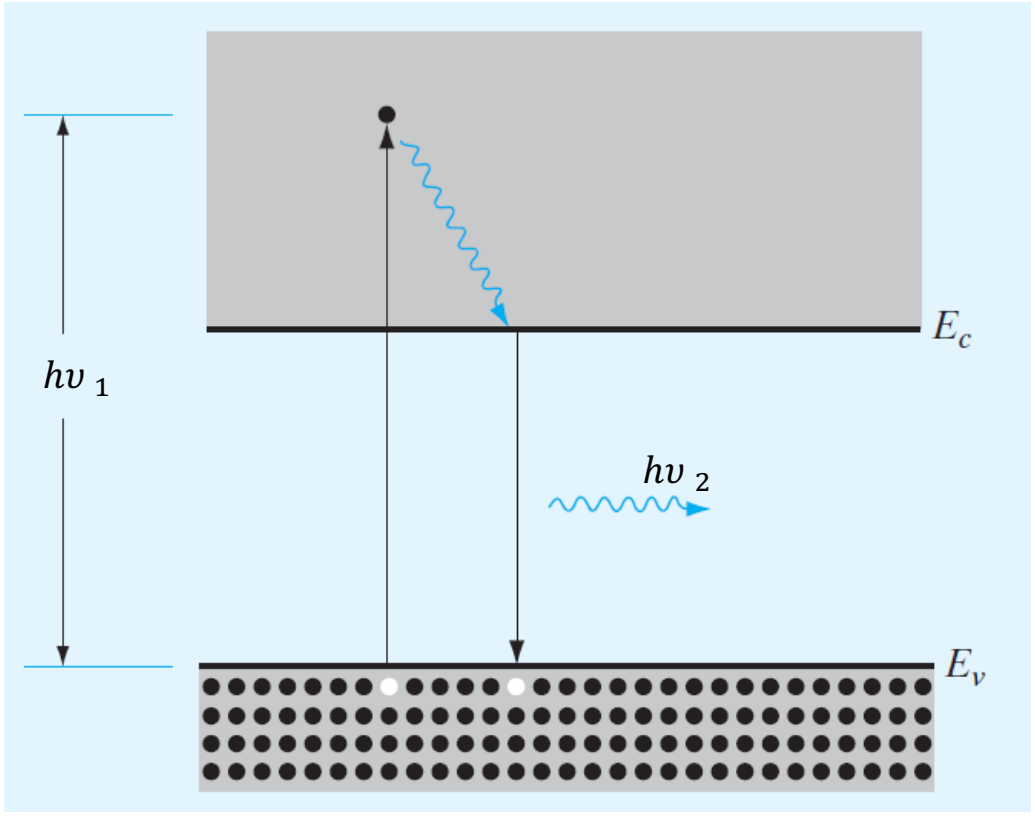


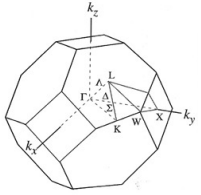
Example

- p-doped GaAs sample with 10^{15} cm^{-3} acceptors. Flash light (on/off) to initially produce $\Delta n = \Delta p = 10^{14}$ EHPs/cm^3 at t=0. Recombination lifetime τ = 10 ns. How do p(t) and n(t) evolve with time?



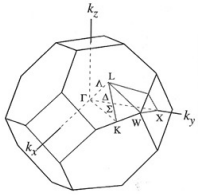
Photoluminescence





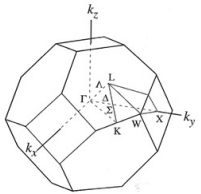
Example

- A sample of Si doped with $N_A = 10^{16}\,cm^{-3}$ boron, with recombination lifetime $\tau = 1\ \mu s$. It is exposed continuously to light, such that electron-hole pairs are generated throughout the sample at the rate of 10^{20} per cm^3 per second, *i.e.* the generation rate $G_L = 10^{20}/cm^3/s$.
- What are equilibrium n_0 and p_0 (before light is on)? $n_0 = 10^4$. $p_0 = 10^{16}$
- How many extra δn and δp are there with light on? $\delta n = \delta p = 2.9 = 10^{14}/cm^3$
- What are the total carrier concentrations with light on?
- What is the np product?



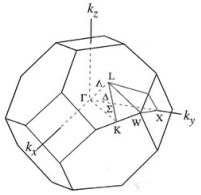
Quasi-Fermi Levels

- Fermi level (E_F) has been an *equilibrium* quantity which gave us n and p
  - $n_0 = n_i \exp\left[\frac{(E_F - E_{Fi})}{kT}\right]$
  - $p_0 = n_i \exp\left[\frac{-(E_F - E_{Fi})}{kT}\right]$
- Q: What does the Fermi level look like when we have excess carriers (from light), and hence *non-equilibrium*?
- So far, we know $np \neq n_i^2$
- We like similar (easy) equations, so we define *quasi-Fermi* levels E_{Fn} and E_{Fp} :
  - $n = n_i \exp\left[\frac{(E_{Fn} - E_{Fi})}{kT}\right]$
  - $p = n_i \exp\left[\frac{-(E_{Fp} - E_{Fi})}{kT}\right]$



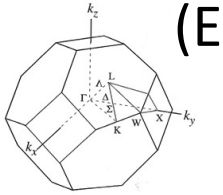
Example

- A sample of Si doped with $N_A = 10^{16}\ cm^{-3}$ boron, with recombination lifetime $\tau = 1\ \mu s$. It is exposed continuously to light, such that electron-hole pairs are generated throughout the sample at the rate of 10^{20} per cm^3 per second, *i.e.* the generation rate $G_L = 10^{20}/cm^3/s$.
- Calculate and draw quasi-Fermi levels from this example.



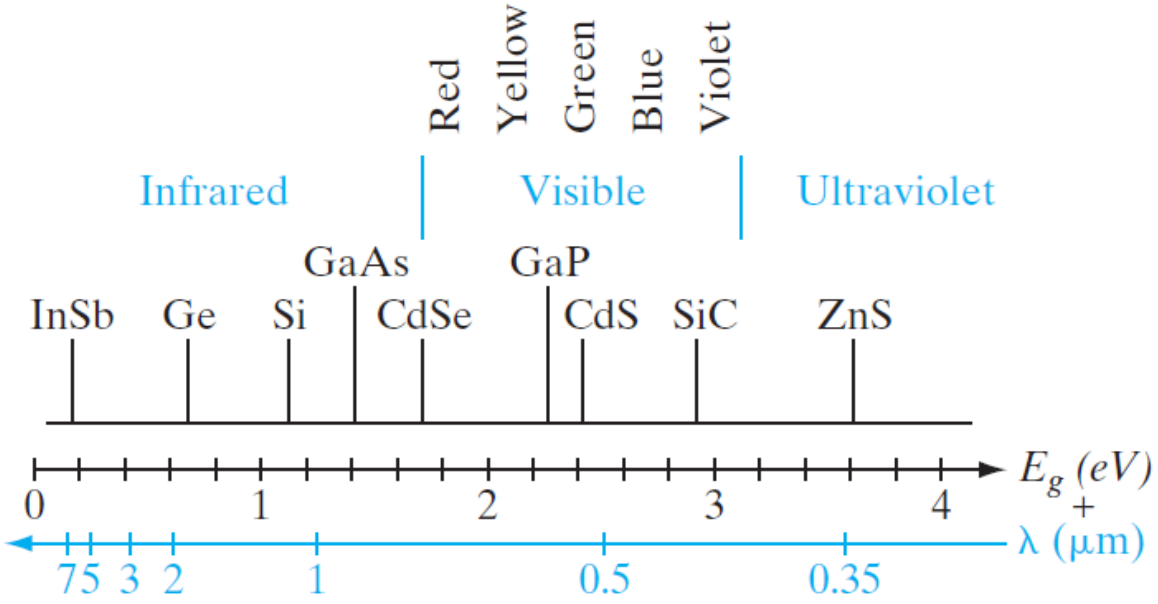
Conductivity (resistivity) Change

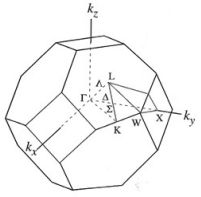
- We have all these excess carriers with the lights ON. Does the conductivity (resistivity) change?
  - Remember: $\sigma = q(\mu_n n_0 + \mu_p p_0)$
- Often before, with lights off, we could neglect the contribution of the minority carriers if the sample was reasonably doped n- or p-type.
- But with lights ON, we have extra carriers δn and δp such that n and p are affected:
- Photoconductivity = change in conductivity due to excess carriers (EHPs) from lights being turned on:



Photoconductive Devices

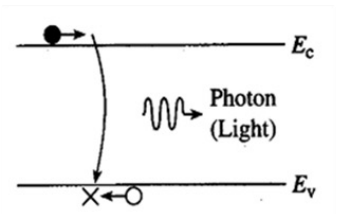
- $\sigma = q(\mu_n n_0 + \mu_p p_0)$
- $\delta n = \tau_n G_{op}$
- $\delta p = \tau_p G_{op}$
- $\Delta\sigma = qG_{op}(\tau_n\mu_n+\tau_p\mu_p)$



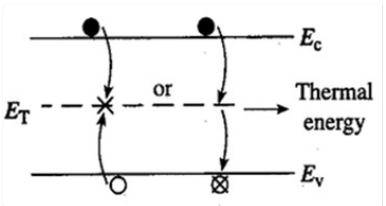


More Recombination Process

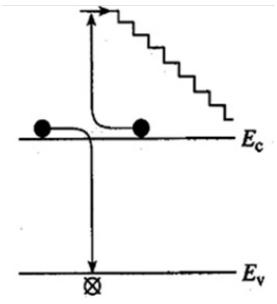
Direct

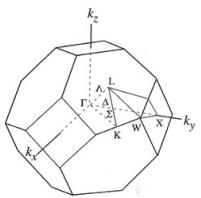


R-G Center



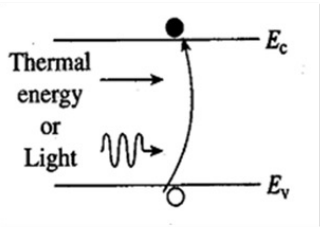
Auger



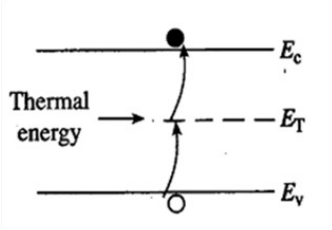


More Generation Process

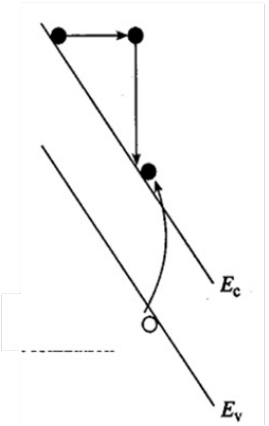
Band-to-Band

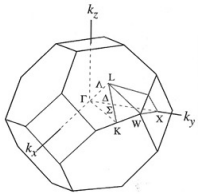


R-G Center



Impact Ionization





Shockley-Read-Hall (SRH) Recombination

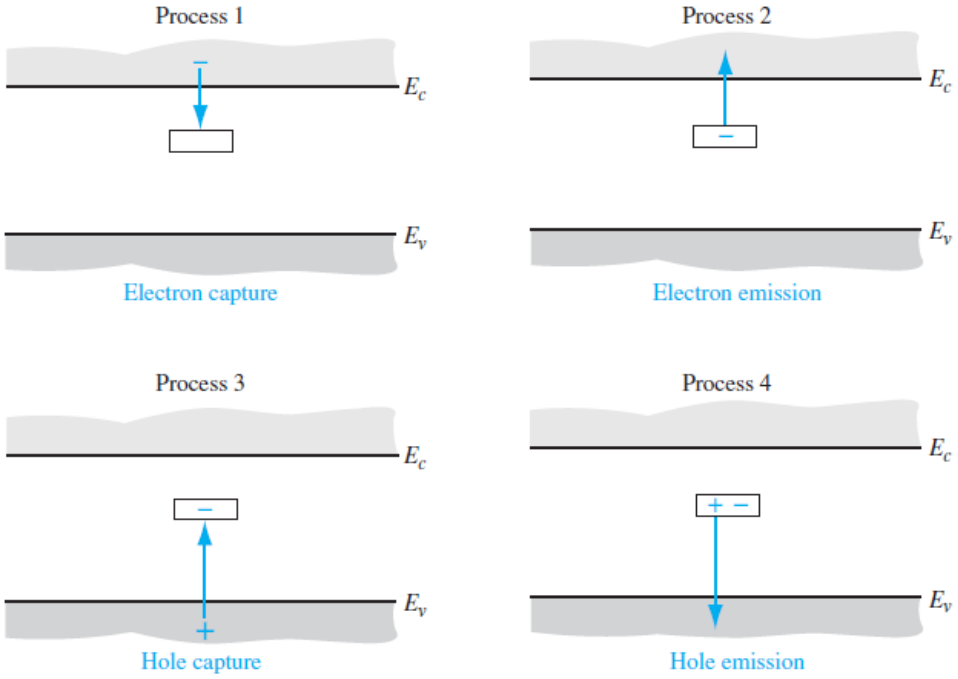
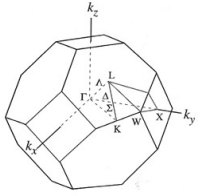


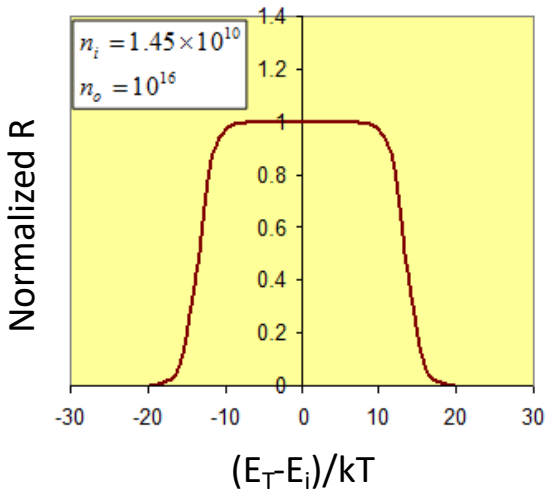
Figure 6.16 | The four basic trapping and emission processes for the case of an acceptor-type trap.

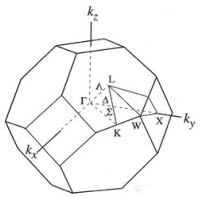


Steady State Net Recombination Rate R

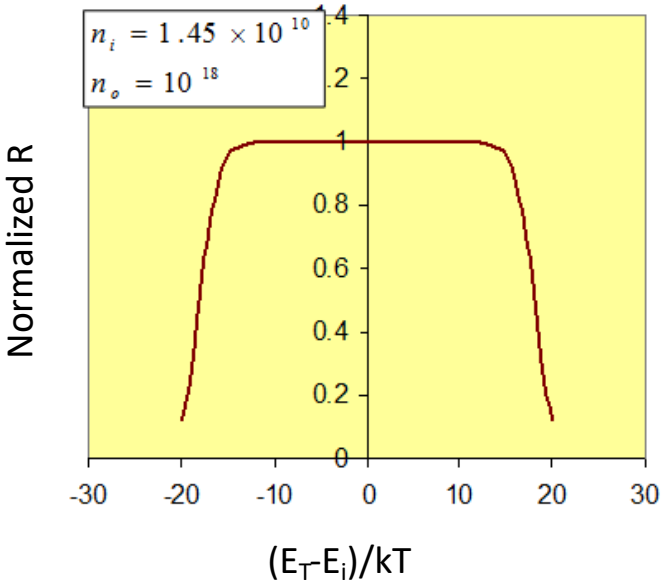
- $R = \frac{(pn - n_i^2)}{\tau_p\left(n + n_i \exp\left(\frac{E_T - E_i}{kT}\right)\right) + \tau_n\left(p + n_i \exp\left(\frac{E_i - E_T}{kT}\right)\right)}$
- Assume $\tau_n = \tau_p$

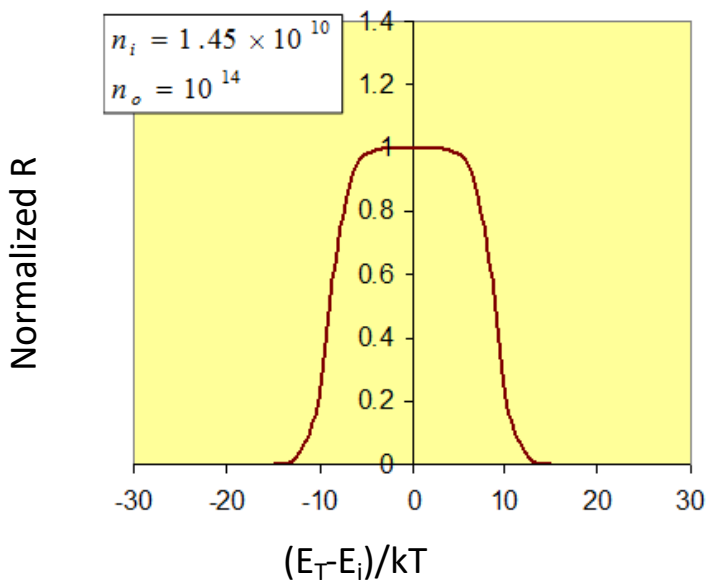
R is large for E_T close to E_i and then drops off rapidly.
Traps near mid-gap are much more efficient as recombination centers.

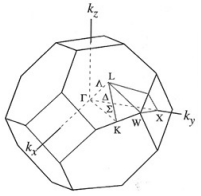




Dependence of Recombination on Doping





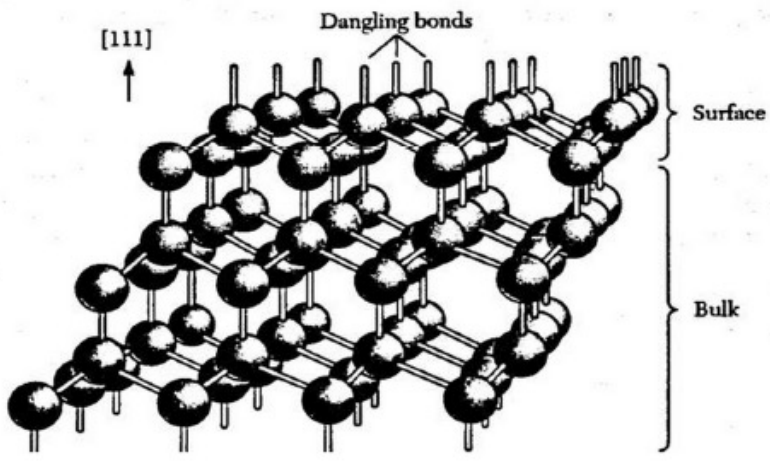


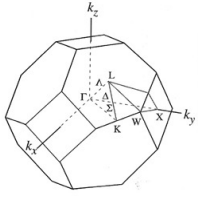
The lighter the doping, the more effective the mid-gap traps compared to the traps near the band edges.

Surface Defects

- Perfect periodic nature ends abruptly
- Periodic potential function is disturbed
- Additional allowed energy states within bandgap
- Surface generation-recombination

Unsatisfied bonds = defects





Surface States

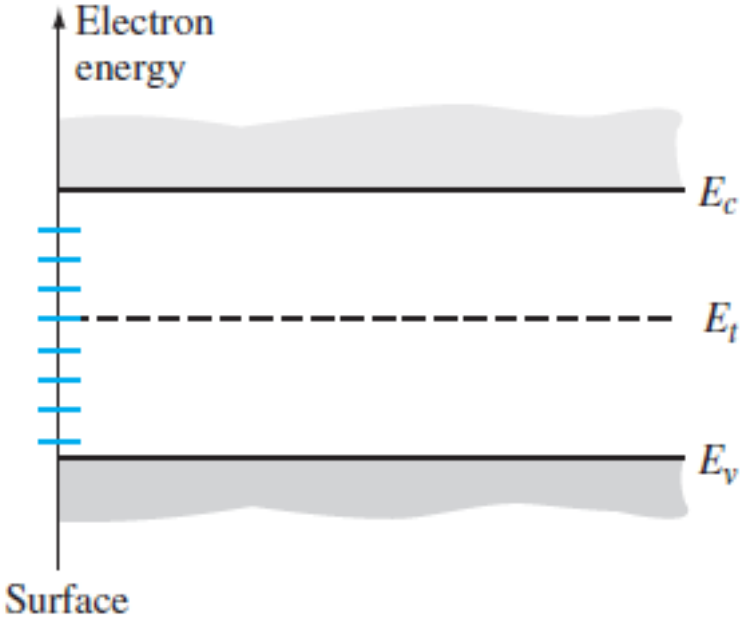
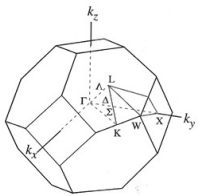
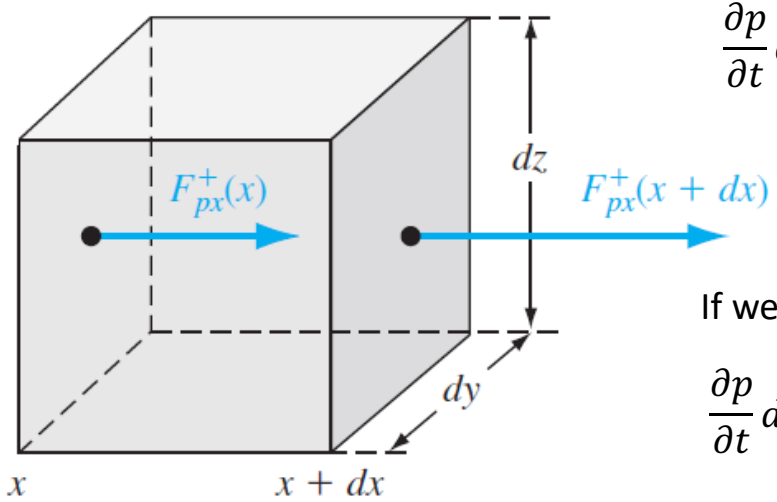


Figure 6.17 | Distribution of surface states within the forbidden bandgap.



Continuity Equation

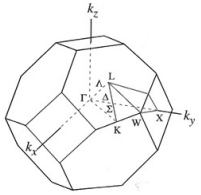


$$\frac{\partial p}{\partial t}dxdydz = \left[F_{px}^{+}(x) - F_{px}^{+}(x+dx)\right]dydz = -\frac{\partial F_{px}^{+}}{\partial x}dxdydz$$

If we include the generation and recombination:

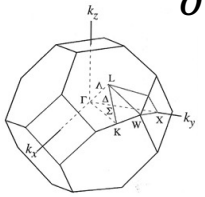
$$\frac{\partial p}{\partial t}dxdydz = -\frac{\partial F_{px}^{+}}{\partial x}dxdydz + g_p dxdydz - r_p dxdydz$$

$$\frac{\partial p}{\partial t} = -\frac{\partial F_{px}^{+}}{\partial x} + g_p - r_p$$



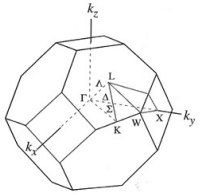
Particle Flux

- $F_{px}^{+} = \frac{J_p}{+e} = \mu_p pE - D_p \frac{\partial p}{\partial x}$
- $F_{nx}^{+} = \frac{J_n}{-e} = -\mu_n nE - D_n \frac{\partial n}{\partial x}$
- $\frac{\partial p}{\partial t} = -\frac{1}{e}\frac{\partial J_p}{\partial x} + g_p - r_p$
- $\frac{\partial n}{\partial t} = +\frac{1}{e}\frac{\partial J_n}{\partial x} + g_n - r_n$
- $\frac{\partial p}{\partial t} = -\mu_p \frac{\partial pE}{\partial x} + D_p \frac{\partial^2 p}{\partial x^2} + g_p - r_p$
- $\frac{\partial n}{\partial t} = +\mu_n \frac{\partial nE}{\partial x} + D_n \frac{\partial^2 n}{\partial x^2} + g_n - r_n$



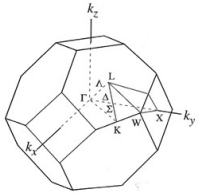
Summary of Transport Equations in 3D

- $\frac{\partial p}{\partial t} = -\frac{1}{q}\nabla \cdot J_q + g_p - r_p$
- $\frac{\partial n}{\partial t} = \frac{1}{q}\nabla \cdot J_n + g_n - r_n$
- $J_p = ep\mu_p \vec{E} - eD_p \nabla p$
- $J_n = en\mu_n \vec{E} + eD_n \nabla n$
- $\nabla \cdot D = q(p - n + N_d^+ - N_a^-)$



Example

- P type semiconductor with “low-level” injection (not in equilibrium)
- Track minority carrier:



Example

Objective: Determine the steady-state spatial dependence of the excess carrier concentration.

Consider a p-type semiconductor that is homogeneous and infinite in extent. Assume a zero applied electric field. For a one-dimensional crystal, assume that excess carriers are being generated at $x = 0$ only, as indicated in Figure 6.6. The excess carriers being generated at $x = 0$ will begin diffusing in both the $+x$ and $-x$ directions. Calculate the steady-state excess carrier concentration as a function of x.

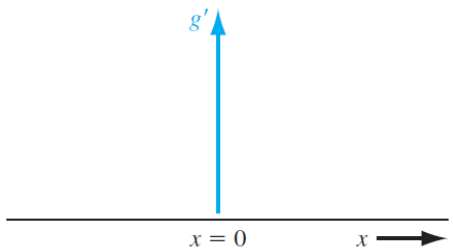


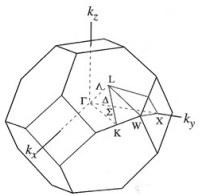
Figure 6.6 | Steady-state generation rate at $x = 0$.

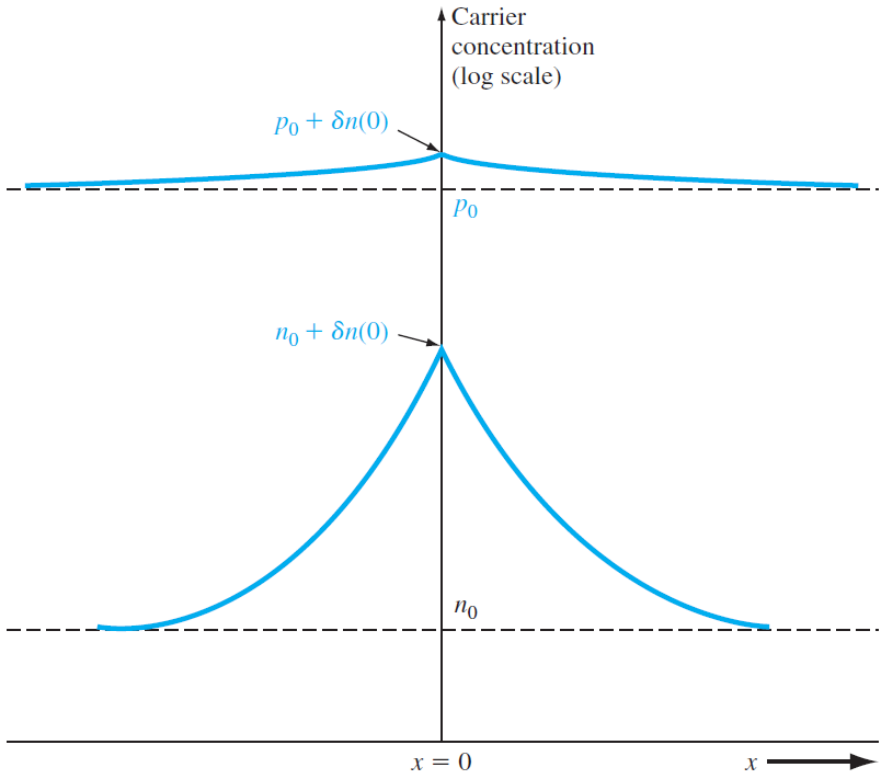
$$\delta n = Ae^{-\frac{x}{L_n}} + Be^{\frac{x}{L_n}} \quad x>0$$

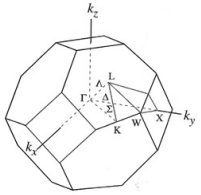
$$\delta n = Ce^{-\frac{x}{L_n}} + De^{\frac{x}{L_n}} \quad x<0$$

$$\delta n(\pm\infty) = 0 \Rightarrow B = C = 0$$

$$\delta n_p(0+) = \delta n_p(0-) \Rightarrow A = D$$




Carrier
concentration
(log scale)
$p_0 + \delta n(0)$
p_0
$n_0 + \delta n(0)$
n_0
$x = 0$
x


k_z
k_y
k_x
L
Λ
Γ
Δ
Σ
K
W
X